

Bridging Cognitive Biases and Fact-Checking Models: A Survey

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Abstract

A new pandemic, formless yet inherently deceptive, has emerged in the digital age, the pandemic of misinformation. Misinformation spreads through digital networks, thriving on human perception and social interaction complexities. Recent advances in misinformation detection signal a key shift toward understanding the psychological and social bases that drive its propagation. This shift underscores the importance of human behaviour in the fight against misinformation, suggesting that simply automating fact-checking is not enough. This survey investigates the evolving relationship between different fact-checking models and psychological concepts, specifically how human cognitive biases, social dynamics, and emotional responses interact with misinformation. We examine various cutting-edge fact-checking models through the prism of social behaviour and human psychology to get fresh insights into their effectiveness and limitations. Furthermore, we offer future research avenues for developing more resilient and adaptable frameworks, such as neuro-behavioural models that consider technology factors while including the intricacies of human cognition and social influence. These techniques can potentially improve disinformation identification, reducing the negative impact of misinformation on society.

1 Introduction

The digital age has fundamentally transformed the way information is disseminated, leading to the rapid and widespread propagation of misinformation [Amoruso *et al.*, 2020]. This phenomenon is more than just the existence of incorrect information; it also entails complex relationships between the information and the entities that consume it. As individuals navigate this complex network of information, their perceptions and behaviours are shaped by a number of psychological and social influences [Ecker *et al.*, 2022].

To tackle misinformation, traditional automated fact-checking approaches draw parallels to the theory of the multi-store memory model from psychology [Atkinson and Shiffrin, 1968]. These approaches involve retrieving relevant

information related to the claim (long-term memory), identifying supporting evidence (short-term memory), and using this evidence to classify the claim (working memory). Historically, misinformation was primarily classified into various categories, including fake news, misleading information, fabricated content and similar other forms [Altay *et al.*, 2023]. However, these classifications often overlooked that accurate information can also contribute to mass misinformation. For instance, consider the statement, ‘A doctor dies after receiving the second dose of a vaccine.’ While factually correct, the timing and societal perceptions surrounding it created a wave of misinformation [Grady and Mazzei, 2021]. Similarly, satirical news, hyper-partisan reporting, and propaganda are frequently debated as sources of misinformation [Altay *et al.*, 2023]. Addressing such challenges requires a shift in focus from the intention behind the information to its impact. Our study emphasizes the role of perception, cognition, and social psychology in understanding misinformation.

Before the emergence of techniques addressing edge cases of misinformation, earlier studies primarily relied on relational operators to match claims with evidence [Krishna *et al.*, 2021]. These approaches focused heavily on evidence retrieval, as seen in datasets such as FEVER [Thorne *et al.*, 2018] and PolitiFact [Pöldvere *et al.*, 2023], which effectively incorporated evidence to train models. Advancements in tool-augmented Large Language Models (LLMs) have transformed this landscape. Modern datasets can largely bypass the evidence retrieval task and instead focus on nuanced aspects such as ambiguity, perception, attitude, and social energy to address contemporary challenges of misinformation [Song, 2021]. Recent literature highlights the application of LLMs to simulate user reactions based on different user profiles, generating interaction graphs [Wan *et al.*, 2024]. This underscores the growing importance of graph-based algorithms in combating misinformation. The future of misinformation prevention will likely leverage Graph Neural Networks (GNNs) in conjunction with LLMs to tackle misinformation’s social and psychological complexities. With the objective of better addressing the complexities of misinformation in today’s context, recent studies have also started to explore these issues by developing specialized datasets with ambiguous names [Chiang and Lee, 2024] and persuasive contents [Xu *et al.*, 2024]. An important factor in the dissemination of misinformation is communication style. Some re-

cent studies have introduced style-agnostic training methods to minimize the influence of writing styles on the detection of misinformation [Wu *et al.*, 2024].

As the lines between AI models and psychological processes become more hazy, it underscores the need for a critical research area at their intersection. While recent studies in the misinformation domain typically concentrate on developing fact-checking models, the boundaries between human cognition and AI are rapidly vanishing. This evolution is a precursor to more robust computational models capable of leveraging data features and network structures to counter adversaries from multiple fronts, a necessity as misinformation campaigns grow more sophisticated. For instance, Loth *et al.* [2024] highlighted how Generative AI is transforming the landscape by automating the creation of misinformation (text and multimodal), effectively manipulating cognition, perception, and attitudes with minimal cost and unprecedented reach. Studies underscore the psychological impact of these advancements and the pressing need for countermeasures [Alam *et al.*, 2021; Kou *et al.*, 2022].

While many surveys focus on structural approaches or literature reviews of fact-checking models, defining misinformation remains a multifaceted challenge. Every study often adopts distinct hierarchies and nomenclature, leading to fragmented research directions. This paper advocates for analyzing misinformation through its impacts rather than its structural characteristics, considering it as a dynamic gestalt or through its functional consequences. Eldifrawi *et al.* [2024] emphasized methodologies for verifying claim veracity and generating justifications. However, the societal and cognitive ramifications of generative technologies in disseminating misinformation warrant deeper exploration [Loth *et al.*, 2024; Alam *et al.*, 2021]. This shift towards functionalism acknowledges the harmful potential of misinformation in reshaping public cognition and highlights the urgent need for interdisciplinary research [Kou *et al.*, 2022]. This evolution is driven by recognising that humans are not simply programmed machines but conscious beings influenced by various biases and shaped by environmental and internal factors [Keller, 2010]. Addressing these vulnerabilities requires innovative tools integrating technology, social psychology, and cognition insights.

2 Understanding the Gestalt of Lies

A gestalt [Wertheimer and Riezler, 1984] is a concept that suggests an organized whole is greater than the sum of its parts. Misinformation is not merely about verifying the veracity of individual claims, nor is it about assessing the truthfulness of their sum total [Starbird *et al.*, 2019]. Its gestalt extends beyond this, encompassing human perception of the information. This introduces the concepts of perception and qualia [Chalmers, 1995] into the traditional approach of verifying individual claims to determine veracity.

The gestalt nature of misinformation has been a major bottleneck in establishing a clear definition. Cognitive psychology provides models such as the multi-store model of memory [Atkinson and Shiffrin, 1968], which has been compared to traditional fact-checking models. However, these cogni-

tive models fail to capture the broader gestalt of misinformation. Current models attempt to incorporate this gestalt in fragmented ways [Wan *et al.*, 2024; Chiang and Lee, 2024; Xu *et al.*, 2024; Wu *et al.*, 2024], which does not fully resolve the problem but offers a direction for future research.

Through this survey, we will examine how misinformation research has evolved from a structural perspective, such as verifying multiple facts within a claim, to a holistic perspective, where augmented biases in datasets and psychological phenomena are integrated into misinformation detection frameworks.

3 Classic Simplicity, But Trust it Sparingly

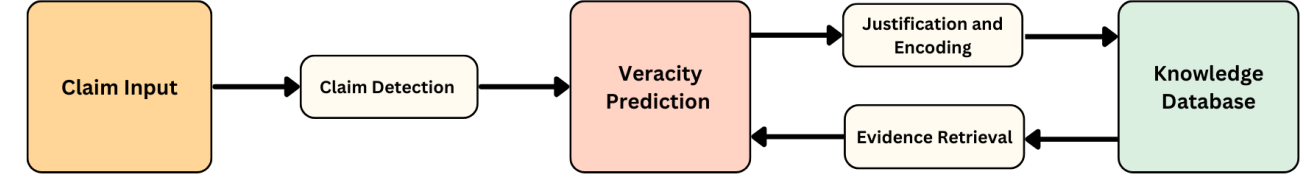
With the rise of the Internet, the problem of misinformation has become increasingly significant. In its early forms, misinformation often involved presenting incorrect facts in a presentable manner. Automated Fact-Checking (AFC) modules were developed to combat this. A recent survey on AFC outlined four general stages involved in these systems [Eldifrawi *et al.*, 2024]. Interestingly, these stages can be compared to the multi-store model of memory [Atkinson and Shiffrin, 1968], which posits that memory consists of three key components: the sensory register (SR), short-term memory (STM), and long-term memory (LTM). Each stage of AFC exhibits similarities to the functions performed by these memory stores, as shown in Figure 1.

Check-worthy Claims. We encounter numerous stimuli daily, but not all require our attention. Only claims that are significant, verifiable, and have the potential to cause harm or mislead should be addressed [Guo *et al.*, 2022; Sundriyal *et al.*, 2023]. The concept of claim check-worthiness can be linked to the process of attention [Atkinson and Shiffrin, 1968], which facilitates the transfer of information from sensory memory to short-term memory in the multi-store memory model. A well-designed attention mechanism ensures that relevant claims are accurately identified and prioritized, enhancing the quality and reliability of the retrieved evidence or information.

Evidence Retrieval. When signals are received, the brain attempts to match them with existing knowledge stored in long-term memory. The retrieved information is akin to evidence that aligns most closely with the claim. Human intelligence is often measured by the efficiency and effectiveness of retrieving the most relevant information. Similarly, the effectiveness of an AFC model is directly proportional to the performance of its evidence retrieval engine [Chen *et al.*, 2024; Sundriyal *et al.*, 2022b]. Various evidence selection approaches have been discussed as classification [Wadden *et al.*, 2020] and regression problem; also, annotation distillation is used to mimic the annotator distribution [Glockner *et al.*, 2024].

Veracity Prediction. Short-term memory holds the most critical information for processing, including sensory input and retrieved knowledge. Similarly, in AFC models, veracity prediction is performed using claims and the evidence that aligns with them. Both short-term memory and the AFC processing modules function as active centres where align-

(A) Automated Fact Checking Pipeline



(B) Multi-Store Model of Memory

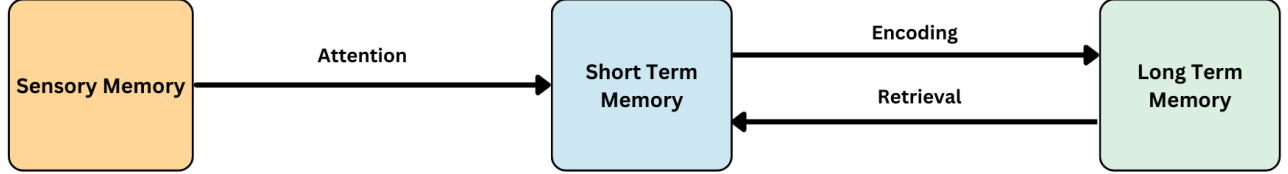


Figure 1: A comparison between (b) Automated Fact Checking Pipeline and (b) Multi-Store Model of Memory, illustrating how misinformation detection parallels human memory processes. The fact-checking pipeline retrieves and verifies claims against a knowledge database, similar to how human cognition processes, stores, and retrieves information.

ment, reasoning, comprehension and classification occur in real-time [Baddeley and Hitch, 1974].

Justification Production. During veracity prediction, information undergoes processing and is organized to generate justifications. This process parallels the memory encoding process, where information is summarized and structured for effective storage [Panigrahy, 2019]. Once encoded, the information can take the form of justifications when shared with users or become part of a knowledge base if stored in memory. Other types of models include relational models such as ProofVer [Krishna *et al.*, 2021] and MultiVers [Wadden *et al.*, 2022]. ProofVer is based on the natural logic theory of compositional entailment, while MultiVers predicts rationale sentences from the evidence and classifies claim veracity using Longformer [Beltagy *et al.*, 2020] as an encoder. These models are highly explainable; however, they face significant challenges when dealing with the context and pragmatics of claims. Additionally, these models do not effectively capture complex relationships between phrases or sentences, limiting their overall performance in certain scenarios.

In the current era of multimodal content, the scope and complexity of misinformation have expanded significantly. Misinformation can encompass advanced techniques of human engineering designed to influence and manipulate large audiences. As a result, evidence-retrieval-based fact-checking methods and relational approaches are increasingly inadequate for addressing the intricacies of modern misinformation. Furthermore, while foundational, the memory model under study oversimplifies and overlooks phenomena related to emotion, bias, cognition and perception [Bennion *et al.*, 2013]. Both models also fail to incorporate human behaviour in response to a claim, instead focusing solely on its content or style. Misinformation is most harmful when it has the potential to influence the masses. Additionally, filtering claims based on their potential to disrupt human behaviour is cost-

effective. This strategy would enhance the efficiency of real-time fact-checking by giving importance to claims that pose large-scale risk. This highlights the need to align our research with recent advancements in psychological studies, providing a more robust framework for understanding and addressing the evolving challenges of misinformation.

In the next section, we will explore how adopting a different point of view can redefine the approach to misinformation, enabling the development of more generalizable, effective, and robust solutions.

4 Claims vs. Humans: Who Defines the Truth?

In the battle against misinformation, a pertinent question is – *Are factual claims or human judgment the ultimate pillars of truth?* Although claims, whether verified or not, provide the foundation for facts, how they are seen and accepted is ultimately determined by human interpretation. Thus, knowing the balance between objective claims and subjective perception becomes critical in establishing what humans believe to be true.

Focus on Claims. Two types of attitude change can be related to perceptions of the veracity of claims: incongruent change and congruent change [McGuire, 1969]. Incongruent change occurs when a claim first believed to be ‘True’ is altered to ‘False’ or vice versa. Congruent change, on the other hand, involves increasing confidence in an existing label. Mathematically, these changes are typically modeled using a general loss function.

According to the psychological literature, incongruent changes are more challenging for humans than congruent changes [McGuire, 1969]. Ranadive *et al.* [2023] highlighted that class-selective neurons tend to emerge within the initial few training epochs. This observation contrasts with the findings of human-based experiments, suggesting that achieving

class label changes would require significantly more effort and extended training. A major limitation of this study includes its focus on using ResNet-50s trained on ImageNet rather than text-based datasets. Extending the analysis to pretrained language models could provide deeper insights into the behaviour and applicability of class-selective neurons across various contexts. One possible explanation for the observed behaviour is the absence of human biases, prejudices, stereotyping, and prewiring in these models, which may make incongruent changes easier. Further research is needed to address this, including developing datasets that can effectively capture these cognitive attributes from language models.

Focus on Humans. Both external and internal factors influence individuals' susceptibility to misinformation and motives to disseminate them. For the former, they are mainly about features of the information environment and related social networks and social groups. For the latter, they refer to the personal traits and cognitive processes, which may play a role. Many meta-analyses have been done to figure out what are the influential psychological factors and individual traits regarding misinformation spreading [Munusamy *et al.*, 2024; Nan *et al.*, 2022; Sultan *et al.*, 2024]. Specific to frequently mentioned cognitive factors, they include familiarity, confirmation bias, heuristic information processing, etc.

As implied by the mere-exposure effect [Zajonc, 1968], individuals may have a positive attitude to the information they come across multiple times. Similarly, familiarity with news is related to a true-news bias [Sultan *et al.*, 2024]. Confirmation bias theory indicates that people may tend to trust more on their favored hypothesis [Klayman, 1995]. Studies further show that people who already have a misperception are more susceptible to the misinformation which aligns with their existing notions, and this, in turn, intensifies polarization in public discussion (which we refer to as congruent change above) [Zhou and Shen, 2022]. The heuristic-systematic model (HSM) posits that people may process information in a heuristic way, which is nonanalytic [Todorov *et al.*, 2002] and makes people more easily to share misinformation [Sun and Xie, 2024].

5 Cognitive and Social Complexities

Several models have sought to break traditional barriers by incorporating human perspectives into their frameworks in recent years. A recent article highlights the challenges of evidence retrieval in real-world scenarios and emphasizes the need for a human-in-the-loop fact-checking system [Chen *et al.*, 2024]. These models aim to address specific limitations of previous AFC systems. Table 1 presents a list of recent studies on fact-checking and misinformation detection and psychological phenomena discussed in them. Following, we discuss some of the major psychological perspectives tackled in recent studies.

Social Energy and Perspectives. LLMs are limited in their ability to be used directly off-the-shelf for judging the veracity of news articles, where factual accuracy is essential. To alleviate this issue, Wan *et al.* [2024] proposed crucial steps in misinformation detection where LLMs may be introduced into the pipeline. Their model generates diverse reactions by

leveraging varied user attributes and creates a user-news network using prompt-based techniques. To simulate the comment propagation process, three distinct strategies are implemented for LLMs: (i) generate a comment based on the news article, (ii) generate a comment in response to an existing comment, and (iii) select a comment for further engagement. The resulting network embodies an artificial social perspective [Song, 2021], which serves as the foundation for performing various tasks using GNNs. This approach integrates user interactions to enhance the model's ability to analyse and interpret complex patterns in misinformation detection. In another model [Ni *et al.*, 2024], debates between LLM agents are used to refine factuality assessments. Through these debates, a reasoned argument is developed, which, when aggregated with step-by-step reasoning for fact extraction, yields enhanced performance.

Despite its innovative approach, these models have certain limitations from a psycho-social perspective. One significant challenge lies in the lack of interpretability in LLMs, making it difficult to verify whether generated reactions are reliable enough for real-world applications. These reactions and reasoning may result from hallucinations or incorporate underlying biases, potentially impacting the robustness of the network. Ensuring the reliability of the network requires metrics to assess whether the generated reactions and reasoning are appropriate and logical, thereby enhancing the model's overall stability and performance. Furthermore, the limited availability of open-source LLMs raises concerns about variability in model behaviour – specifically, how training on different datasets may influence the generation of user perspectives. The prompts designed for the generation of these reactions and reasoning are also subject to human biases, which may limit the range of outcomes. To address these issues, extensive research is required, incorporating real-world datasets and focusing on robust mechanistic interpretability [Elhage *et al.*, 2021]. Such efforts would benefit from greater reliance on open-source models, ensuring transparency and consistency in evaluating and applying these systems.

Oversimplification of Cognitive Bias. Chiang and Lee [2024] highlighted a critical issue in LLMs, where information about multiple entities is mixed within the same biography. A reader without prior knowledge cannot discern that the information pertains to different entities. Furthermore, previous factuality metrics assumed that as long as each claim is factual, the combination of those claims is factual [Min *et al.*, 2023]. This vulnerability is exacerbated by entity ambiguity within datasets. To address this, the authors introduced a new metric, D-FActScore, designed to evaluate better the factuality of paragraphs impacted by entity ambiguity [Chiang and Lee, 2024]. However, the proposed solution represents an oversimplification of cognitive bias, as it addresses the outcomes rather than the underlying causes [Kahneman *et al.*, 1982]. Future research must prioritize the development of efficient evidence-retrieval models that go beyond basic search functionalities. These models should incorporate intelligent heuristics capable of mitigating pitfalls and biases in retrieved information [Chaiken *et al.*, 1989].

Additionally, future studies could explore the role of false

Previous Work	Research Focus	Psy. Study	Psychological Phenomenon
Wu <i>et al.</i> [2024]	Misinformation Detection, Fact-Checking Models	✓	Framing Effect, Cognitive Bias
Xu <i>et al.</i> [2024]	Misinformation Impact, Domain-Specific Techniques	✓	Persuasion, Heuristic-Systematic Model, Confirmation Bias, Attitude-Behaviour Consistency, Cognitive Dissonance
Wan <i>et al.</i> [2024]	Fact-Checking Models, Misinformation Detection	✓	Social Energy
Ni <i>et al.</i> [2024]	Claim Structuring, Misinformation Detection	✓	Social Energy, Groupthink
Chiang and Lee [2024]	Claim Structuring, Misinformation Detection	✓	Cognitive Bias, Availability Heuristics
Saha and Srihari [2024]	Misinformation Detection, Multimodal Techniques	✓	Groupthink, Availability Heuristic
Deng <i>et al.</i> [2024]	Claim Structuring, Domain-Specific Techniques	✗	N/A
Luo <i>et al.</i> [2024]	Claim Structuring, Domain-Specific Techniques	✓	Negativity Bias
Pan <i>et al.</i> [2023]	Fact-Checking Systems, Claim Structuring	✗	N/A
Fajcik <i>et al.</i> [2023]	Fact-Checking Systems, Claim Structuring	✗	N/A
Mendes <i>et al.</i> [2023]	Misinformation Detection, Domain-Specific Techniques	✓	Anchoring Bias
Yue <i>et al.</i> [2023]	Domain-Specific Techniques, Fact-Checking Models	✗	N/A
Liu <i>et al.</i> [2023]	Multimodal Techniques, Claim Structuring	✓	Modality Bias
Gupta <i>et al.</i> [2022]	Multimodal Techniques, Fact-Checking Models	✓	Priming Effect
Thai <i>et al.</i> [2022]	Claim Structuring, Multimodal Techniques	✗	N/A
Wright <i>et al.</i> [2022]	Fact-Checking Models, Domain-Specific Techniques	✗	N/A
Ousidhoum <i>et al.</i> [2022]	Claim Structuring, Fact-Checking Models	✗	N/A
Chen <i>et al.</i> [2022]	Claim Structuring, Misinformation Detection	✗	N/A
Glockner <i>et al.</i> [2022]	Misinformation Impact, Multimodal Techniques	✓	Confirmation Bias
Sundriyal <i>et al.</i> [2022a]	Claim Structuring, Fact-Checking Models	✗	N/A
Jiang and Wilson [2021]	Claim Structuring, Misinformation Detection	✗	N/A
Kazemi <i>et al.</i> [2021]	Domain-Specific Techniques, Claim Structuring	✓	Cultural Bias
Sheng <i>et al.</i> [2021]	Claim Structuring, Fact-Checking Models	✗	N/A
Tymoshenko and Moschitti [2021]	Fact-Checking Systems, Misinformation Detection	✗	N/A
Da <i>et al.</i> [2021]	Multimodal Techniques, Misinformation Impact	✓	Visual Bias, Attribution Error
Schlichtkrull <i>et al.</i> [2021]	Fact-Checking Models, Claim Structuring	✗	N/A

Table 1: Overview of misinformation-related papers categorized by their research focus, with indicators of psychological phenomena studied.

cognates — words that appear similar in two different languages but have different meanings — in spreading misinformation and hate. This area of research holds significant potential for addressing a unique and impactful dimension of the misinformation problem.

LLMs’ Belief Towards Misinformation via Persuasive Conversation. Xu *et al.* [2024] demonstrated that LLMs tend to tailor their responses even to follow an objectively wrong viewpoint. This behaviour parallels human tendencies to resolve cognitive dissonance [Festinger, 1957], where LLMs attempt to reduce the conflict between their initial belief and persuasive input. Another significant finding is that most LLMs are susceptible to persuasive misinformation, particularly when it aligns with their prior knowledge or training data. This susceptibility mirrors confirmation bias in humans [Nickerson, 1998]. Additionally, the repetition strategy significantly increases the misinformation rate across most models. Repetition acts as a heuristic, making misinformation appear more credible without requiring systematic processing, aligning with the Heuristic-Systematic model theory [Chaiken *et al.*, 1989]. Furthermore, LLMs exhibit sycophancy, corresponding to the theory of Attitude-Behaviour

Consistency [Wicker, 1969]. While the literature uncovers valuable insights, it also highlights limitations. In humans, beliefs are shaped by more complex cognitive processes, including emotional investment, experiential memory, and subconscious biases. These more profound layers of human cognition are not addressed in this study.

Humans often resist persuasion through strategies such as counter-arguing, source skepticism, or reliance on prior knowledge [Brehm and Brehm, 2015]. Incorporating these active resistance mechanisms into LLM studies could help develop more robust strategies to counter misinformation effectively.

Style-based Stereotypes. Wu *et al.* [2024] highlighted style manipulation using LLMs as a significant challenge to misinformation detection. It reveals that fake news camouflaged with LLM-generated styles substantially reduces state-of-the-art text-based detectors’ effectiveness. Cognitive bias, a mental shortcut that aids quick situational analysis, is reflected in these models, exhibiting stereotypes toward certain styles when predicting veracity [Kahneman *et al.*, 1982]. Reframed news from Trusted publishers leverages their credibility as a tool for deception, where credibility of the message

Pre-2024	Before Internet Democratization	Fact-checking relied on newspapers and pamphlets, often politically motivated. No formal mechanisms for accuracy verification. Relied on human judgment through heuristics or close contacts' experiences.
	Rise of Social Media	Fact-checking using statistical NLP. Techniques: Relational Entailment and TF-IDF for evidence retrieval. Early models: IBM Watson, Stanford NLP models.
	Data Explosion	Use of deep learning for fact-checking and justification production. Integration of statistical NLP concepts. Models: ProofVer, MultiVerS.
2024	Change in Misinformation Perception	Robust algorithms for fact-checking with focus on addressing psychological pitfalls in LLMs. Psycho-social concepts for enhanced effectiveness in fact-checking. Models: DELL, D-FactScore, SheepDog Persuasive Conversation.
Post-2024	Cognitive Models Creep into AI	Increased research in psychology to understand AI models' behavior. Integration of psychological principles into AI fact-checking systems.
	Mechanistic Interpretability	Interpretability techniques to uncover LLM processes when studying psychological aspects. Improved understanding of "black box" behavior in LLMs.
	Neuro-Behavioral Models	Hybrid GNN and LLMs for enhanced reasoning and adaptability. Incorporating Psychological Principles and ensure transparency and comprehensive insights.

Figure 2: The timeline of the evolution of misinformation research, demonstrating significant advances in fact-checking approaches from earlier times (pre-2024), present (2024), and projected future (post-2024).

source strongly influences compliance and belief [Pornpitakpan, 2004]. Humans evaluate persuasive content using either the central route, which relies on logical reasoning, or the peripheral route, focusing on stylistic cues [Petty and Cacioppo, 1986]. Stylistic manipulation takes advantage of the peripheral route, bypassing critical analysis and enhancing the effectiveness of misinformation.

However, several areas require further research. Emotional influence, combined with stylistic factors, could provide deeper insights. Investigating individual differences in perceiving styles as trustworthy could lead to developing more robust and adaptable models. Additionally, the current evaluation framework focuses solely on individual interactions with content and neglects the social amplification of misinformation. Comparing and contrasting the effects of style and social influence could offer valuable insights into how these factors collectively and interactively shape belief in misinformation [Song, 2021].

6 Future Scope and Directions

The findings discussed in this work provide an excellent basis for future research in the field of misinformation, as illustrated in Figure 2. However, several topics still require further investigation. This section covers prospective future research directions, emphasising expanding current procedures and exploring new and novel ways.

Datasets Unlocking Psychological Biases. The discussed literature falls within this domain, highlighting the limita-

tions of existing datasets, such as FARM [Xu *et al.*, 2024] and AmbigBio [Chiang and Lee, 2024], which are not designed to address multiple biases simultaneously. Psychological biases, perceptions, emotions, and motivations have intricate and interdependent relationships strongly influenced by the environment in which they occur. Consequently, there is a pressing need to develop datasets capable of tackling multiple psychological pitfalls concurrently. Creating such datasets will require extensive text annotations, necessitating collaboration between experts in linguistics and psychology. These datasets could enable the training of advanced models, including LLMs, to recognize and address these biases effectively. However, the impact of training LLMs on such datasets on their downstream task performance remains uncertain. Investigating this aspect could provide valuable insights into the development of artificial general intelligence. This might also facilitate the downgrading of certain capabilities or offer moral and emotional reasoning in LLMs, ensuring their efficiency and making them safer to deploy.

Mechanistic interpretability offers a promising avenue for progress by modeling weight changes and the latent architecture of these models [Elhage *et al.*, 2021]. This could help integrate psychological aspects into LLMs without the need for additional datasets, thereby reducing the reliance on human labour and saving significant time. Such advancements would pave the way for more nuanced, ethical, and effective AI systems.

Data with Demographic Depth. One of the significant challenges in analysing misinformation from a human-centric

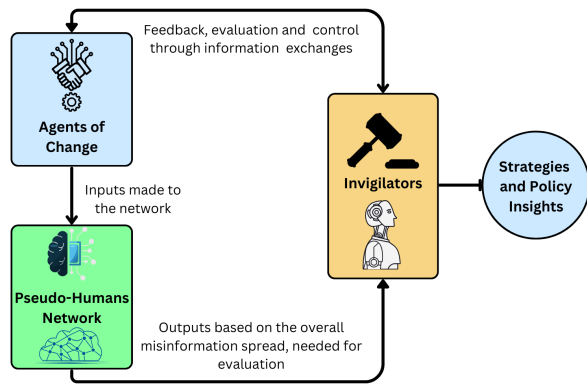


Figure 3: An illustration of the neuro-behavioural model framework. The figure illustrates the interaction dynamics among the three main components of the model: agents of change, pseudo-humans network, and invigilators.

perspective is the unavailability of social media network data that includes both content and demographic information. This challenge arises due to data regulations, privacy, and security issues. Access to such datasets would enable the analysis of inter and intra-group interactions given a specific piece of multimedia content. Such an analysis could facilitate the development of an initial classifier capable of identifying data with potential misinformation and devising targeted solutions to address it. By incorporating temporal aspects, such as timestamps and historical patterns of misinformation spread, predicting specific periods of vulnerability and identifying groups that may be more susceptible during those times becomes possible. This analysis is particularly suited for state-based organizations rather than private entities. Using appropriate abstraction methods, these datasets could be securely shared with designated state agencies to combat misinformation effectively while safeguarding privacy and ensuring ethical use.

Neuro-Behavioural Models. World models have recently gained significant attention due to their ability to simulate the interaction between an agent and its environment [Ha and Schmidhuber, 2018]. These models create a latent representation of the world, enabling them to predict environmental dynamics by integrating perception, memory, and decision-making. By simulating scenarios internally, world models allow agents to evaluate potential actions without direct interaction, making them effective for planning and problem-solving. Multi-agent systems are essential for addressing social problems, as they interact not only with the environment but also with one another [Guo *et al.*, 2024]. Neuro-Behavioural Models conceptualize social interactions within a simulated world. These models feature three primary types of agents, as illustrated in Figure 3.

1. **Pseudo-humans:** Large models that simulate human-like attributes, including perception, biases, and cognitive frameworks. These models can incorporate varying proportions of biases to ensure diversity.
2. **Agents of Change:** Models interacting with pseudo-humans, providing inputs and analyzing outputs. They

simulate numerous scenarios to study interactions.

3. **Invigilators:** Models that continuously evaluate the network and provide feedback to the Agents of Change, enabling dynamic input adjustments. After analysis, they assist in developing new policies and strategies.

This high-level framework requires advanced techniques, such as GNN, Reinforcement Learning, and their integration with LLMs. With further advancements in world model frameworks, Neuro-Behavioural Models could unlock new possibilities for simulating and addressing complex social challenges.

7 Conclusion

This survey reviewed the evolution of methods used for misinformation prevention, emphasizing the need for a paradigm shift to address the challenges of combating misinformation in the modern era. With adversaries leveraging LLMs and the increasing complexity of online media interactions, focusing on the receiver rather than solely on the source of misinformation is crucial. The discussion highlighted how various models embody psychological concepts related to memory, cognition, biases, and persuasion. Future advancements in misinformation research require progress in three key areas: psychology, mechanistic interpretability, and integrating GNNs with LLMs to explore the interplay of innate and learned behaviours in preventing misinformation spread. The lack of comprehensive network data paired with textual counterparts often results in the reliance on synthetic data, which has yet to be validated for reliability. The four studies explored in this survey emphasize integrating social perspectives [Wan *et al.*, 2024], developing robust fake news detection methods [Wu *et al.*, 2024], influencing LLM beliefs through persuasion [Xu *et al.*, 2024], and introducing a metric for scoring ambiguous names [Chiang and Lee, 2024], highlighting deeper analysis from a human-engineering perspective.

A significant focus on the impact of misinformation, rather than its source or textual characteristics, arises from the absence of a universally accepted definition of misinformation. Decades of psychological research on human behaviour provide a foundation for understanding how different population segments respond to misinformation. These insights can be integrated with existing deep learning frameworks, such as LLMs and GNNs, to move beyond traditional pattern matching towards neuro-behavioural models. By incorporating cognitive and behavioural science principles, future models could simulate and predict human responses to misinformation by analyzing internal cognitive processes and external social stimuli. Such predictive capabilities hold transformative potential, including preventing conflicts, conserving resources, protecting the environment, and mitigating social issues where human behaviour is central. Moreover, these advancements could pave the way for real-time intervention strategies, enabling personalized misinformation resistance mechanisms tailored to different cognitive profiles. This fusion of AI and behavioural science could redefine misinformation research, offering a more adaptive, context-aware, and human-centric approach to addressing one of the most pressing challenges of the digital age.

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